Small centralizers and the rank-four obstruction for Ree groups

Human Compression project: recovery of L039

This paper supplies a replacement for the rank-four argument of Leemans, Schulte and Van Maldeghem [1]. It uses their classical subgroup setting and Pierro's centralizer-intersection theorem, rather than the normality assertion in their final rank-four argument. The previously reviewed companion *Involution blocks and generation in the small Ree groups* supplies the complete rank-three construction, the exclusion of proper almost simple extensions, and the upper bound of four. Its full proof remains an acquired part of the recovery. Neither the subgroup classifications nor the theory of Ree groups are being reproved here.

1 Elementary obstruction lemmas

All actions of automorphisms are on the right: $x^{\alpha\beta} = (x^\alpha)^\beta$, and $x^{\mathrm{inn}(u)} = u^{-1}xu$. A string marking $(r_0, \ldots, r_{d-1})$ consists of independent involutions with $[r_i, r_j] = 1$ for $|i-j| > 1$ and the intersection property

$$\langle r_i : i \in I\rangle \cap \langle r_i : i \in J\rangle = \langle r_i : i \in I \cap J\rangle.$$

Independence follows from this property and the requirement that every generator is nonidentity. In a nonabelian simple group, adjacent generators cannot commute: a commuting adjacent pair would cut the string into two nontrivial commuting normal subgroups.

Lemma 1 (Dihedral product obstruction)**.** *Let E be an elementary abelian 2-group and let D_{2n} be dihedral of order $2n$, with n odd. No subgroup of $E \times D_{2n}$ has a rank-three string marking (x, y, z) with both xy and yz of order greater than two.*

Proof. An involution projects either to the identity or to a reflection in D_{2n}. A trivial projection would make it central, so all three projections are reflections. Since x, z commute, their reflections coincide: two distinct reflections in an odd dihedral group do not commute. Write $xy = (e, r)$ and $yz = (f, r^{-1})$, where r has odd order $d > 1$. Squaring removes the central 2-coordinates, and r^2 generates $\langle r\rangle$. Thus both $\langle x, y\rangle$ and $\langle y, z\rangle$ contain $\{1\} \times \langle r\rangle$. This nontrivial odd-order subgroup cannot be contained in $\langle y\rangle$, contradicting the intersection property. □

Lemma 2 (An end centralizer)**.** *Let a centerless group G be generated by a string (a, b, c, d) of involutions, with c, d noncommuting and $\mathrm{C}_G(c, d)$ abelian. If $K = \langle b, c, d\rangle$ has an involution in its centralizer in G, then such a marking is impossible.*

Proof. If t is that involution, then $a, t \in \mathrm{C}_G(c, d)$, so a commutes with t. The other three generators already commute with t. Consequently $t \in Z(G)$, contrary to the hypothesis. □

Lemma 3 (The field-degree reduction)**.** *Let $G = \langle a, b, c, d\rangle$ be as above, and suppose $\mathrm{C}_G(c, d)$ is a subgroup of a Klein four-group. If a nonidentity automorphism τ of odd order fixes b, c, d, then $|\tau| = 3$. Moreover $\mathrm{C}_G(c, d)$ is a Klein four-group on whose nonidentity elements τ acts cyclically.*

Proof. The centralizer is τ-invariant and contains a. If τ fixed a, it would fix every generator, hence be the identity. Its induced action is therefore a nontrivial odd-order automorphism of a subgroup of a Klein four-group. The centralizer must be the full four-group, and the induced action has order three. Hence τ^3 fixes a and b, c, d, so is the identity on G. This proves both assertions. □

Theorem 4 (Two conjugate fixed subgroups)**.** *Let G be a finite centerless group with elementary abelian Sylow 2-subgroups. Assume that the centralizer of each pair of noncommuting involutions is either elementary abelian of characteristic three or a subgroup of a Klein four-group. There is no rank-four string marking (a, b, c, d) with noncommuting adjacent pairs for which*

$$K = \langle b, c, d\rangle = \mathrm{Fix}(\tau), \qquad L = \langle a, b, c\rangle = \mathrm{Fix}(\sigma),$$

where τ has order three and σ has the same image as τ in $\mathrm{Aut}(G)/\mathrm{Inn}(G)$.

Proof. There is a unique $u \in G$ with $\sigma = \tau\,\mathrm{inn}(u)$. Both automorphisms fix b, c, hence $u \in A := \mathrm{C}_G(b, c)$. If $u = 1$ the two fixed subgroups coincide, contrary to independence; thus $u \neq 1$. The group A is τ-invariant because τ fixes b, c. The intersection property gives $K \cap L = D := \langle b, c\rangle$.

First suppose A is elementary abelian of characteristic three. The nontrivial group $U = \langle u, u^\tau, u^{\tau^2}\rangle$ is τ-invariant. An automorphism of order dividing three of a nontrivial finite 3-group has a nonidentity fixed point: its orbits have lengths one or three, so the number of fixed points is divisible by three. Choose $1 \neq v \in U$ fixed by τ. Since $U \leq A$ is abelian, v commutes with u, and is therefore fixed by σ as well. Thus $v \in K \cap L = D$, while $v \in A = \mathrm{C}_G(D)$. This puts an element of order three in $Z(D)$, impossible for a dihedral group, whose center has order at most two.

Otherwise A is a 2-group of exponent two, so $u^2 = 1$. Since $a^\sigma = a$, we have $a^\tau = a^u$. The elements a, a^τ are distinct because $a \notin K$. Both centralize c, d, and hence commute. Their four-group $V = \langle a, a^\tau\rangle$ is normalized by u, which interchanges its two displayed generators. Consequently $\langle V, u\rangle$ is a nonabelian 2-group of order eight. This contradicts the elementary abelian Sylow 2-subgroups of G. □

2 The retained Ree structure

Write $R(q) = {}^2G_2(q)$ for $q = 3^f$, f odd, and first assume $q \geq 27$. The following inputs are retained explicitly from the classical description summarized in [1, Section 2.2] and [2, Section 2 and Lemma 3.13].

1. The group $G = R(q)$ is centerless and simple, its Sylow 2-subgroups are elementary abelian of order eight, and $\mathrm{Aut}(G)/G$ is cyclic of order f.

2. Its maximal subgroups are proper subfield Ree groups, point stabilizers of order $q^3(q-1)$, involution centralizers $C_2 \times \mathrm{PSL}_2(q)$, four-group normalizers $(C_2^2 \times D_{(q+1)/2}) : C_3$, and the two cyclic Hall normalizers $C_{q+1\pm\sqrt{3q}} : C_6$. In the four-group normalizer the indicated C_3 acts nontrivially on the four-group. Here the subscript of D always denotes its order.

3. For noncommuting involutions x, y, the group $\mathrm{C}_G(x, y)$ is elementary abelian of characteristic three or a subgroup of a Klein four-group. This is precisely the noncommuting case of Pierro's Lemma 3.13; the commuting case has a larger centralizer.

4. A standard subfield subgroup $R(s)$, where $s = 3^h$ and $h \mid f$, is the fixed subgroup of the corresponding field automorphism of order f/h. Every subgroup of this subfield type is conjugate to the standard one. This fixed-field identification is part of the standard matrix realization of the Ree groups; see Kleidman [3, Lemma 6.2], as explicitly restated in [4, Section 6, p. 7]. The automorphism acts by the indicated field map on matrix entries. For $s = 3$ its fixed subgroup is $R(3) = \mathrm{PSL}_2(8) : C_3$, whose derived group is $\mathrm{PSL}_2(8)$.

5. The group $\mathrm{PSL}_2(8)$ has maximal subgroups $C_2^3 : C_7$, D_{14} and D_{18}. Its nonidentity element orders are $2, 3, 7, 9$. Its split tori of order seven and their dihedral normalizers form single conjugacy classes. In $R(3)$ the outer element acts as the field automorphism of $\mathrm{PSL}_2(8)$. These are the usual degree-eight instance of the subgroup and automorphism descriptions for PSL_2 used in [1].

6. Every element of order three or nine fixes exactly one point in the natural action of $R(q)$ of degree q^3+1. The point stabilizer has 2-part two. These facts are recorded in [2, Tables 1 and 3].

These inputs do not include the disputed normality assertion or the middle-dihedral assertion of source Lemma 3.13. No finite computation is a premise of the arguments below.

Proposition 5 (Possible end parabolics)**.** *In a hypothetical rank-four string marking of $G = R(q)$, each end parabolic is a subfield $R(s)$ with $s \geq 27$, or the derived group $\mathrm{PSL}_2(8)$ of a subfield $R(3)$.*

Proof. Consider $K = \langle b, c, d\rangle$. It is proper, has the intersection property, and its adjacent pairs do not commute. Choose a maximal subgroup containing it. Point stabilizers and cyclic Hall normalizers have Sylow 2-subgroups of order two, whereas $\langle b, d\rangle$ has order four. An involution centralizer is impossible by Lemma 2. In a four-group normalizer, every involution lies in the normal subgroup $C_2^2 \times D_{(q+1)/2}$ of index three. Write $(q+1)/2 = 2n$; since $q \equiv 3 \pmod 8$, n is odd. Lemma 1 excludes this case.

Only a subfield Ree group remains. Repeat this argument within a proper subfield as long as K is proper there. An involution centralizing K at any stage still lies in G, so the same exclusion applies. The strictly decreasing subfield orders make the process terminate, either with $K = R(s)$, $s \geq 27$, or with $K \leq R(3)$. In the latter case all its involutions lie in $R(3)' = \mathrm{PSL}_2(8)$. A proper subgroup of this last group lies in one of its listed maximal subgroups. The involutions in $C_2^3 : C_7$ all lie in the elementary abelian normal subgroup and cannot give a noncommuting adjacent pair. The odd dihedral groups D_{14} and D_{18} cannot contain two distinct commuting involutions. Hence $K = \mathrm{PSL}_2(8)$. The dual argument applies to $L = \langle a, b, c\rangle$. □

Proposition 6 (Reduction to a cubic field extension)**.** *If an end parabolic is $R(s)$, $s \geq 27$, then $q = s^3$. If it is $\mathrm{PSL}_2(8)$, then $q = 27$.*

Proof. Choose the corresponding field automorphism τ fixing that end parabolic pointwise. In the first case its fixed subgroup is $R(s)$; in the second it is a conjugate of $R(3)$ and its order is f. This automorphism is nonidentity because its fixed subgroup is proper in the simple group G. The centralizer of the last adjacent pair contains the first generator, an involution, so the characteristic-three alternative is impossible and it is a subgroup of a four-group. Lemma 3 gives $|\tau| = 3$, proving the two conclusions. □

3 The two final cases

If $q > 27$, Proposition 6 excludes an end parabolic of type $\mathrm{PSL}_2(8)$. Both ends are therefore $R(s)$ with $q = s^3$, and hence are conjugate. Let τ be the field automorphism fixing one end. Conjugating τ by an inner automorphism gives an automorphism σ whose fixed subgroup is the other end and whose image in $\mathrm{Aut}(G)/G$ is the same. Theorem 4 gives a contradiction. This argument replaces all three parts of source Lemma 3.13 and its use in Lemma 3.14; in particular it does not assume the asserted restriction on the middle dihedral order.

It remains to treat $q = 27$. Both end parabolics must now be $\mathrm{PSL}_2(8)$. We give the ordered-triple argument in the full automorphism group, where the required transitivity holds.

Lemma 7. *In this exceptional case, $|cd| = |ab| = 7$.*

Proof. The possible orders are $3, 7, 9$. If cd had order three or nine, its unique fixed point in the natural Ree action would be fixed by both c and a: conjugation by c inverts cd, whereas a centralizes it. The four-group $\langle a, c\rangle$ would then lie in a point stabilizer of 2-part two, a contradiction. Thus $|cd| = 7$. The dual argument gives $|ab| = 7$. □

Proposition 8 (The exceptional automorphism obstruction)**.** *The group $R(27)$ has no rank-four string marking.*

Proof. For an ordered triple (x, y, z) occurring here, $D = \langle y, z \rangle$ is D_{14} inside a subfield derived group $K \cong \mathrm{PSL}_2(8)$, and $1 \neq x \in \mathrm{C}_G(D)$. All such subfield derived groups are conjugate; their D_{14} subgroups are conjugate inside them. Moreover the normalizer of D in $R(3)$ induces the full automorphism group of D_{14}. To see the last assertion explicitly, use the split torus $T = \langle r \rangle$ of order seven in $\mathrm{SL}_2(8)$ and its basis-swapping involution w. Inner conjugation by T and w induces the translations and inversion on the seven reflections wr^i. The field map on $\mathbb{F}_8$ induces multiplication by two on their indices. Since $\langle -1, 2 \rangle = \mathbb{F}_7^\times$, these actions give all affine permutations $i \mapsto h + ki$ and are transitive on ordered pairs of distinct reflections. This element of order three in $R(3)$ is separate from the following outer automorphism of G.

Fixing such an ordered pair (y, z), let τ be the order-three field automorphism of G fixing its containing $R(3)$ pointwise. No nonidentity involution of $\mathrm{C}_G(D)$ is fixed by τ: any such involution would belong to $R(3)$, hence to K, whereas $\mathrm{C}_K(D) = 1$. The latter equality is immediate in the displayed 2-dimensional model: an element centralizing the split torus is diagonal and commutation with w forces it to be the identity. The centralizer is a subgroup of a four-group and contains x; it is therefore the full four-group, and τ cycles its three nonidentity elements. Combining these actions proves transitivity of $\mathrm{Aut}(G)$ on the relevant ordered triples.

Apply this to (a, c, d) and (d, b, a), which satisfy the same conditions by Lemma 7. Some automorphism interchanges a and d, and thus induces a transposition on the nonidentity elements of $V = \langle a, d \rangle$. However $\mathrm{N}_G(V)$ induces only C_3 on V, by the retained four-group-normalizer structure. In $\mathrm{Aut}(G)$ the quotient $\mathrm{N}_{\mathrm{Aut}(G)}(V)/\mathrm{N}_G(V)$ has odd order, since it embeds in $\mathrm{Aut}(G)/G$. Its induced group on V therefore has odd order as well and cannot contain a transposition. This is a contradiction. □

Theorem 9 (Complete principal conclusion)**.** *For a simple small Ree group $R(q)$ with $q \geq 27$, the only rank of a string C-group representation is three. Every such $R(q)$ has a representation of rank three. No proper almost simple extension of $R(q)$ is generated by involutions, and $R(3)$ is not generated by involutions.*

Proof. The propositions and Theorem 4 exclude rank four. The previously accepted companion proves the bound of four, the existence in rank three, and both odd-quotient exclusions. Ranks one and two generate cyclic or dihedral groups and are impossible for a nonabelian simple group. Together these statements give exactly the principal conclusion of [1, Theorem 1.1]. □

The source does not enumerate all rank-three markings, and neither does this recovery. The theorem is a recovery of the published existence and exclusion statements; no new historical priority, conjecture answer, complete formalization or numerical factor is claimed.

4 Auxiliary claims and the subgroup diagram

The preceding proof deliberately avoids most normalizer calculations in the source. We record their precise status so that this change of route does not hide a source claim. In particular, an implication whose hypothesis is a rank-four Ree marking is discharged by the nonexistence theorem; it is not thereby asserted as an unconditional subgroup-normalizer theorem.

Dihedral and projective-line calculations. In $D_{2n} = \langle r, w \rangle$, a subgroup of order $2m$, $m > 1$ and $m \mid n$, is $\langle r^{n/m}, wr^j \rangle$. A rotation r^i normalizes it exactly when $n/m \mid 2i$; multiplying by its reflection gives the other half of the normalizer. The normalizer therefore has order $2m$ if n/m is odd and $4m$ if it is even. This proves preprint Lemma 1, published Lemma 2.1.

For $L = \mathrm{PSL}_2(q)$, $q = 3^f$ with f odd, we retain the standard split and nonsplit torus description: their orders are $(q-1)/2$ and $(q+1)/2$, their normalizers are dihedral of twice those orders, and a nonidentity semisimple element of order greater than two has its torus as centralizer. Nontrivial unipotents have order three. A dihedral subgroup D_{2d}, $d \geq 3$, cannot have $d = 3$: conjugation inverting a nontrivial unipotent, after its unique projective fixed point is moved to

infinity, would require an upper triangular determinant-one matrix whose diagonal ratio is -1. That ratio is a square, whereas -1 is not a square for $q \equiv 3 \pmod 4$. Thus its rotation lies in a torus, and $2d \mid q-1$ or $q+1$. In particular $4 \nmid d$, and even d can only occur in the latter case. Its normalizer lies in the normalizer of its rotation torus. The preceding dihedral calculation determines it. This is the complete calculation needed in source Lemma 3.5.

For clarity about Lemma 3.6, let $J = D_{2d} \le C_2 \times L$ with $d \ge 3$. Project onto L, obtaining a nonabelian dihedral group D. Either $J = C_2 \times D$, or J is the graph of a homomorphism $\chi : D \to C_2$. In the graph case,

$$N_{C_2 \times L}(J) = C_2 \times \{g \in N_L(D) : \chi(x^g) = \chi(x) \text{ for all } x \in D\}.$$

If the rotation order of D is odd, its only possible nontrivial homomorphism to C_2 distinguishes rotations and reflections and is preserved by every automorphism. If it is even, $|N_L(D) : D|$ cannot be two: that would give a subgroup of L with order divisible by eight, whereas $|L|$ has 2-part four. Inner automorphisms preserve χ. Consequently, in every case

$$N_{C_2 \times L}(J) = C_2 \times N_L(D).$$

When $|D| = 2d$, the latter normalizer is $C_2 \times D_{2d}$, except when $2d \mid q+1$ and $(q+1)/(2d)$ is even, in which case it is $C_2 \times D_{4d}$. When $J = C_2 \times D$, we have $|D| = d$, $d/2$ odd, and $N_L(D) = D_{2d}$, so the result is again $C_2 \times D_{2d}$. These formulas prove the embedding-independence assertion in Lemma 3.6(d). The "according as" order in its part (a) interchanges the two signs; the containing torus normalizer has order $q-1$ when $2d \mid q-1$, and $q+1$ when $2d \mid q+1$.

Source Lemma 3.4 is immediate from the intersection property: r_0 centralizes G_{01}, but if it normalized the proper nontrivial G_0 then G_0 would be normal in the simple generated group. The preceding arguments prove the end-parabolic form actually needed from Lemmas 3.7 and 3.10 directly, with its noncommuting-adjacency hypothesis explicit. To recover the nondegenerate string version of the independent subgroup statement in Lemma 3.7, one may stop the descent at $C_2 \times \mathrm{PSL}_2(s)$ instead of excluding it. Dickson's retained subgroup list at odd powers of three consists of subfield groups, odd-order point stabilizers, the two torus normalizers, and the $\mathrm{PSL}_2(3) = A_4$ endpoint. The first three reductions work as above. In $C_2 \times A_4$, all involutions generate an elementary abelian group, so cannot supply the required adjacent pairs. Finally a subgroup projecting onto a simple $\mathrm{PSL}_2(s)$ is either that factor or its direct product with C_2, since a simple nonabelian group has no nontrivial homomorphism to C_2. This gives the four source families under the nondegenerate string convention used in its proof. The literal claim without that convention is too broad: $C_2^3 \le R(q)$ itself has a rank-three string C-group marking, with adjacent labels two.

For source Lemma 3.8, all involutions of $R(3)$ lie in $K = \mathrm{PSL}_2(8)$. The product of two distinct involutions has order $3, 7$ or 9. Each corresponding dihedral group D has odd rotation order and is self-normalizing in K: use the split torus of order seven or the nonsplit torus of order nine and the dihedral normalizer calculation (including its order-six subgroup). Thus every involution normalizing D already lies in D, which is exactly the asserted $N^0_{R(3)}(D) = D$.

The conjugacy-class diagram. Figure 1 is interpreted as in the source: its boxes are specific conjugacy classes, not every subgroup of the abstract isomorphism type. It applies when $3 \mid f$, so that a subfield $H = R(27)$ exists. Put $J = R(3)$, $K = J'$, and let $D = D_{14}$ be a dihedral subgroup of K. Put $N = N_G(D)$. We retain the standard subfield conjugacy and normalizer data used for this diagram:

$$N_G(H) = H, \quad N_G(J) = J, \quad N_G(K) = J, \quad N = (C_2^2 \times D) : C_3, \quad |N| = 168.$$

The subfield and normal-dihedral data are recorded in [2, Lemmas 2.5, 2.7, 2.9–2.10]; the $K = J'$ normalizer is also part of the source's subfield description. The group N contains four subgroups

abstractly isomorphic to D_{14}, but only one belongs to the displayed class, namely its normal factor D. Indeed the other three are obtained by multiplying its reflections by a fixed nonidentity element of the four-group. An element of order three cycles these three twists; their normalizers have order 56, rather than 168, so they are not G-conjugate to D. To compute those normalizers, note that any normalizing element preserves the characteristic rotation subgroup of order seven. In its torus normalizer a rotation must square into that subgroup, and the C_3-factor cannot preserve a nonzero four-group twist. This gives exactly $C_2^2 \times D_{14}$. The unique normal D_{14} is characteristic in N, whence $N_G(N) = N_G(D) = N$.

All subfield and normal-dihedral classes in question have the same fusion in H as specified above. Also $|J| = 1512$, $|K| = 504$, and $N_K(D) = D$, so K contains $504/14 = 36$ members of the displayed dihedral class. The class sizes and all edge multiplicities in the figure now follow by double counting incident subgroup pairs:

upper and lower classes	lower per upper	upper per lower
G, H	$\|G\|/\|H\|$	1
H, J	$\|H\|/1512$	1
J, K	1	1
K, D	36	4
H, N	$\|H\|/168$	1
N, D	1	1

The five proper class sizes are respectively $|G|/|H|$, $|G|/1512$, $|G|/1512$, $|G|/168$, and $|G|/168$. For example the number of overgroups K of a fixed D is $(|G|/1512)36/(|G|/168) = 4$. No enumeration of all subgroup classes is being claimed. The rank-four replacement does not depend on the uniqueness assertions read from this diagram.

Edition and correction boundary. The published eighteen-page primary was inspected through its publisher-hosted PDF text representation; its exact PDF bytes were not available to the local downloader. The locally bound preprint remains the byte-pinned primary. Published Lemmas 3.12–3.14 retain the proof steps at issue. In Lemma 3.14, the asserted normality of $H = D : C_3$ inside $N = (C_2^2 \times D) : C_3$ would imply that $H/D = C_3$ is normal in $N/D = A_4$, which it is not. The old normality inference is therefore not a premise here. Likewise, for a fixed ordered generating pair of D_{14}, the inner stabilizer is its four-group centralizer and cannot cycle that centralizer's three nonidentity elements. Proposition 8 uses the full automorphism group instead. Finally, the last generation sentence of source Lemma 3.15 must use the third involution outside the torus normalizer, as proved in the companion, not the first involution already chosen inside it.

The source's question about chiral polytopes of higher rank is kept as an unanswered source-recorded question. We make no current-status or priority claim for that question. All source implications assuming a rank-four Ree marking are settled by Theorem 9; their intermediate normalizer reasoning is superseded, rather than silently certified.

References

[1] D. Leemans, E. Schulte and H. Van Maldeghem, *Groups of Ree type in characteristic 3 acting on polytopes*, Ars Math. Contemp. **14** (2018), 209–226, `https://cage.ugent.be/~hvm/artikels/235.pdf`. The local preprint is arXiv:1501.01125v1, eighteen pages; its Lemmas 12–14 correspond to published Lemmas 3.12–3.14.

[2] E. Pierro, *The Möbius function of the small Ree groups*, Australas. J. Combin. **66**(2) (2016), 142–176, `https://ajc.maths.uq.edu.au/pdf/66/ajc_v66_p142.pdf`.

[3] P. B. Kleidman, *The maximal subgroups of the Chevalley groups $G_2(q)$ with q odd, the Ree groups ${}^2G_2(q)$, and their automorphism groups*, J. Algebra **117** (1988), 30–71.

[4] G. P. Nagy, *Embeddings of Ree unitals in a projective plane over a field*, Finite Fields Appl. **74** (2021), 101875. The field-centralizer statement is explicitly restated in Section 6, p. 7 of arXiv:2007.10464v3, `https://real.mtak.hu/156937/1/2007.10464.pdf`.